ModelIt!

Student Activity Pack

Does the Sun Warm Everything the Same?

Sunlight, Shade, and Temperature

Kindergarten | NGSS K-PS3-1 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Circle which spot is warmer: (picture of a sunny sidewalk / picture of a shady sidewalk)

2 Draw the sun warming something up.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Amount of Sunlight

Surface Temperature

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Shady Spot

Amount of Sunlight set to low (shade)

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Sunny Spot

Amount of Sunlight set to high (full sun)

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Does the sun warm a sunny spot and a shady spot the same way? Use evidence from your simulation to support your answers.

8 STEM Challenge: Design a Shade Shelter

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YOUR ENGINEERING MISSION

Build a small shelter that blocks sunlight and keeps a toy animal cool on a hot sunny day.

THE SCENARIO

It is a super hot day at the zoo, and the toy animals need to cool down! Your team must build a shelter that creates the most shade so the animals do not get too warm. Use what you learned about sunlight and shade to design the best shelter!

Engineering Design Thinking

1. WHAT MATERIALS WILL YOU USE TO BLOCK THE SUN?
2. HOW BIG DOES YOUR SHELTER NEED TO BE TO COVER THE TOY ANIMAL?
3. HOW WILL YOU TEST IF YOUR SHELTER REALLY KEEPS THINGS COOLER?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Solar Energy Scientists study how sunlight can be used to power homes and cars. They help us use the sun's energy in amazing ways! They earn $70,000-$120,000/year.